

LightCGS-Net: A novel lightweight road extraction method for remote sensing images combining global semantics and spatial details

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ABSTRACT

Road extraction remains a critical area of study within the realm of remote sensing image. The simultaneous extraction of accurate global semantic and spatial detail features from such images poses a significant challenge in contemporary research. This paper introduced a novel road extraction methodology, termed LightCGS-Net, which integrated a spatial detail branch derived from a lightweight CNN and a global semantic branch based on an enhanced Swin Transformer within the encoder. The spatial detail branch enhances the accuracy of tiny road extractions, while the global semantic branch preserves comprehensive contextual information. To address the noise interference caused by the extensive redundant data produced by the dual-branch fusion, Gaussian filtering was employed. The Lightweight Parallel Channel and Space Attention Mechanism (PCSAM) in the skip connection addresses road discontinuities caused by tree and building obstructions. The Feature Fusion Mechanism (FFM) module, using dilated convolution, captures edge features and provides precise structural road information, with distinct loss functions for primary and edge roads. Testing on the DeepGlobe, Massachusetts, and SpaceNet 03-05 datasets showed that our network performed well compared to traditional road extraction networks, surpassing them in evaluation scores and segmentation effectiveness, despite having more parameters and computational time. LightCGS-Net stands out in road networks using Transformer technology due to its lower parameter (43.78M) and flops (196.09G). It performs well in IoU metrics and maintains competitiveness in Recall, F1-score, and Accuracy. It shows superior generalization and exceeds traditional semantic segmentation methods in overall effectiveness.

1. Introduction

In addition to serving as the foundation of contemporary urban infrastructure and geographic information systems (GIS) (Church, 2002), roads are essential components of modern transportation systems. Road data (Wu et al., 2014) were vital to electronic navigation maps since they are basic components of geographic information databases. Furthermore, in military operations, route information is a vital goal. Forces can use flying aircraft to quickly collect road network data in new areas or precisely object roadways to impede traffic. Furthermore, the unique topology of road networks can facilitate auxiliary location and object search operations when the wireless positioning systems of military UAVs experience interference.

High-resolution remote sensing images (HRSIs) (Wu et al., 2014; Sun et al., 2022; Chen et al., 2022a) provide exceptional spatial resolution and detailed surface information, making them indispensable in

fields such as earthquake rescue, traffic planning, urban development, agricultural monitoring, and disaster assessment. They have become a crucial data source for national strategic planning. However, the geometric complexity and dense road networks in these images can lead to spatial information loss, hindering accurate road extraction. Shadows from trees and buildings may further obscure road surfaces, causing data gaps and discontinuities, and in some cases, shadows may be mistakenly identified as roads. Traditional road extraction methods (Wang et al., 2016a; Lian et al., 2020), which rely on manually designed techniques based on spectral, geometric, color, texture, and topological features, as well as clustering and classification algorithms, are resource-intensive and often yield suboptimal results. These methods also struggle to accurately capture global semantic features like road connectivity and network density, and fail to extract fine spatial details such as road edges, widths, and intricate shapes. Therefore, the challenge remains to develop techniques that can effectively capture

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both global semantic context and fine spatial details in road feature extraction from HRSIs.

The advent of deep learning has revolutionized road extraction in HRSIs. Convolutional Neural Networks (CNNs) (Li et al., 2021), with their rapid advancements and exceptional performance, have become the foundation of computer-aided road extraction. U-Net (Ronneberger et al., 2015), a seminal model for semantic segmentation, stands out for its U-shaped architecture and skip connections, facilitating feature segmentation through upsampling and downsampling. Consequently, many road extraction networks, such as DeepLabV3 (Yurtkulu et al., 2019) and D-LinkNet (Zhou et al., 2018), are built on U-Net. DeepLabV3 leverages dilated convolutions to maintain spatial pixel relationships and uses spatial pyramid pooling to aggregate road features, addressing the challenges of high-resolution image road extraction. D-LinkNet, a prominent network, employs an encoder-decoder architecture with cascaded dilated convolutions to extend the receptive field, enabling extraction of high-level features while refining boundaries using low-level spatial details. However, limitations in receptive field expansion can lead to over-dilation, reducing its effectiveness. Given these constraints, CNN-based methods alone struggle to balance global semantic extraction with precise road boundary detail preservation. Achieving effective information complementarity between roads is crucial to breaking through accuracy limitations.

Inspired by the success of Transformers in natural language processing (NLP) (Chowdhary and Chowdhary, 2020b), the emergence of the Vision Transformer (ViT) (Khan et al., 2022) marks the successful introduction of the Transformer architecture into the field of computer vision. ViT leverages the self-attention mechanism to deeply model contextual information within sequences, proving highly effective in extracting road features from remote image across various spectral bands. However, while ViT performs well in road feature extraction tasks for large-scale remote sensing or multi-spectral images, its Multi-head Self-Attention (MSA) mechanism (Voita et al., 2019) introduces significant computational load and overhead, making it unsuitable for deployment on portable mobile edge computing devices. Liu et al. (2021) proposed a new ViT, called Swin Transformer (ST), that capably serves as a general-purpose backbone for computer vision. Challenges in adapting Transformer from language to vision arise from differences between the two domains. These qualities make it compatible with a broad range of vision tasks. Additionally, Yang et al. (2021) proposed a road extraction method for HRSIs that combines transformers network (TransRoadNet). It provides foreground context information through different stages of the decoder, which can improve the inference ability for occluded areas. However, the integration of Transformers in this model still results in slow processing times for road segmentation in HRSIs, necessitating further optimization to enable faster inference on mobile devices such as UAVs.

To solve these problems, we propose a dual-branch parallel downsampling method within an encoder-decoder segmentation network to improve global and local feature fusion in HRSIs. Each image is resized to half its original dimensions via bilinear interpolation. The full-resolution image feeds into a global semantic branch with a lightweight ST block, utilizing self-attention for global context feature extraction. Simultaneously, the downsampled image enters a spatial detail branch that replaces conventional 2D convolutional Res blocks with depth-wise convolution (DW-Conv) (Guo et al., 2019) and point-wise convolution (PW-Conv) (Hua et al., 2018), ensuring high feature extraction accuracy while minimizing parameters and computational cost. A skip connection between the global semantic branch's first layer and the decoder's final layer introduces a multi-scale Feature Fusion Module (FFM) to enhance edge detail extraction. Element-wise addition is applied to the spatial detail branch's downsampled features, with dilated convolutions reducing noise from redundant information. A second skip connection links the spatial detail branch to the decoder, enabling cascaded operations. This includes a parallel channel-spatial attention mechanism (PCSAM) for spatial mapping from coarse to fine resolution,

effectively resolving road discontinuities caused by occlusions. The parallel structure and lightweight design reduce both model complexity and inference time. This is a brief overview of our most significant contributions:

- (1). We propose a dual-branch road extraction network in the encoder, combining parallel spatial detail and global semantic branches. It innovatively integrates lightweight CNN and Transformer.
- (2). The skip connection incorporates PCSAM and FFM components with feature enhancement capabilities to address challenges in HRSIs edge detail extraction and road discontinuities caused by tree or building occlusions.
- (3). We designed the modules (ST block, CNN, and PCSAM) for lightweight processing, optimizing extraction accuracy while preserving efficient image processing and inference speed.

2. Related works

2.1. Deep-learning-based road extraction methods

In the broad realm of computer vision, road feature extraction (Wang et al., 2016b) has consistently played a crucial role. With the rapid advancement of deep learning technologies, cutting-edge models such as CNNs and Transformers have become essential tools for improving the accuracy of road feature extraction. These models continuously expand the boundaries of road extraction, enhancing the performance and applicability of recognizing and extracting road features in complex environments. They provide novel approaches to address various challenges in practical applications.

CNNs have demonstrated their powerful capabilities in road extraction tasks. The Fully Convolutional Network (FCN) (Long et al., 2015) pioneered a novel application of CNNs in image segmentation by employing skip connections to seamlessly integrate feature maps from different layers, significantly enhancing the network's ability to capture image details. This approach is vital for augmenting the model's ability to capture intricate details. In extracting roads from high-resolution remote images, combining an encoder with multi-scale features and employing dilated convolution techniques broadens the model's receptive field, thereby improving its proficiency in understanding and segmenting complex road structures. Liu et al. (2024) proposed an innovative end-to-end road segmentation approach that effectively mitigates the potential imbalance between CNN depth and spatial resolution by leveraging the multi-level characteristics of convolutional layers. Wang et al. (2021) introduced non-local operations as a generic framework to capture these dependencies, formulating responses at specific positions as the weighted sum of features across all locations, culminating in NL-LinkNet, which leverages differentiable non-local operations for modeling long-range interactions. Moreover, to facilitate deployment on mobile devices, prior research in this field has introduced various lightweight CNNs. SqueezeNet (Koonce and Koonce, 2021) proposed five modules, replacing some 3×3 convolution kernels with 1×1 kernels. MobileNetV1 (Prasetyo et al., 2022) introduced DW-Conv, which decomposes standard convolution into DW-Conv and PW-Conv, significantly reducing parameters. MobileNetXt incorporated the sandglass block, mitigating information loss by inverting the bottleneck structure. Hu et al. (2023) proposed a location-guided network (LGN) for promoting connectivity performance in a very effective and efficient way.

Compared to CNNs, which focus on local features, the attention mechanism in Transformers enables learning of long-range dependencies and global information, making them a crucial area of interest in computer vision. Zeng et al. (2024) proposed a method for road extraction from HRSIs by integrating high-level semantic features with contextual foreground information to improve accuracy. Jiang et al. (2022) developed a pyramid-based ViT tailored for road extraction, where images are partitioned into non-overlapping patches with positional encoding, embedded as sequences, and processed through a

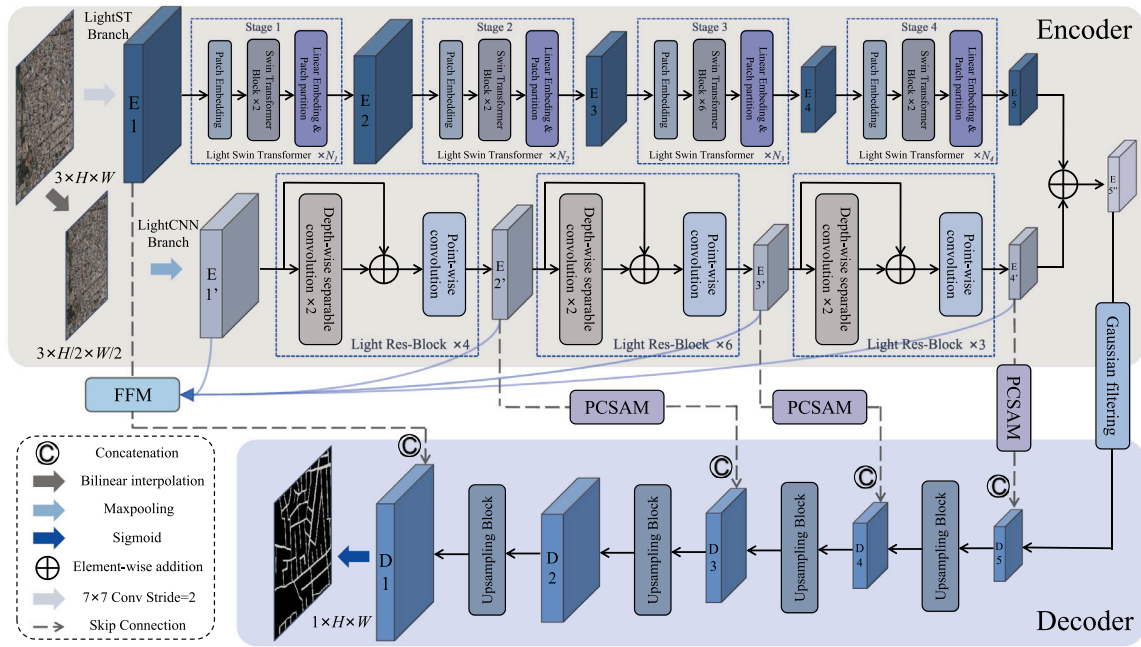


Fig. 1. Overall architecture of the proposed LightCGS-Net for road extraction. Our framework consists of the following three parts: (1) A dual-input branch designed for the simultaneous capture of both global semantic and spatial detail features of roads. (2) PCSAM is employed to capture more complex spatial information. (3) FFM for edge feature enhancement. Moreover, the purpose of Gaussian filtering is to eliminate noise interference caused by repeated and redundant information after dual-branch fusion.

standard Transformer model. Cao et al. (2022) proposed Swin-Unet, which is an Unet-like pure Transformer for medical image segmentation. The tokenized image patches are fed into the Transformer-based U-shaped Encoder-Decoder architecture with skip-connections for local-global semantic feature learning. Effectively addressing numerous challenges in the medical domain. Chen et al. (2022) proposed TransUNet, which merits both Transformers and U-Net, as a strong alternative for medical image segmentation. On one hand, the Transformer encodes tokenized image patches from a CNN feature map as the input sequence for extracting global contexts. On the other hand, the decoder upsamples the encoded features which are then combined with the high-resolution CNN feature maps to enable precise localization. Yuan et al. (2024) introduced a network called SCTransNet, a spatial channel cross-transformer network, which addresses these challenges by employing spatial channel cross-transformer blocks on top of the skip connections in a U-shaped architecture.

Our work is to establish a novel dual-branch network architecture over road feature extraction, in which the decoder side applies ST-based global semantic branching and CNN-based spatial detail branching in parallel. The global and detail features obtained from downsampling the two paths are subsequently combined as the encoder's final layer. Additionally, low computational expenses and global-local feature capture capability were simultaneously assured with the lightCNN and lightST designs.

2.2. Enhanced feature extraction methods for complex background

In road extraction tasks using HRSIs, it is essential to manage road feature details at multiple scales. Unlike other geographic features such as buildings, vegetation, and lakes, roads typically exhibit a linear form. Therefore, focusing on the spatial context of background image is crucial, encompassing the dependencies between geographic entities and their backgrounds, and correlations among similar features. Networks designed to capture long-range dependencies can significantly improve road semantic segmentation performance.

To effectively establish topological relationships between occluded road segments and ensure continuity in road extraction, current research primarily utilizes multi-scale features, feature pyramid pooling,

and dilated convolutions to capture contextual information. Zhao et al. (2021) proposed a spectral Cascaded Dilated Convolutional Network (CDCNet) designed to obtain a larger receptive field for long-ranged information and extract multi-scale features. Subsequently, a Parallel Multipath Network is proposed to capture large-scale, middle-scale, and small-scale spatial features in parallel during all three stages. As PMN deepens in different stages, more multi-scale information flows into the network, enabling finer classification results. Lin et al. (2017) proposed the inherent multi-scale, pyramidal hierarchy of deep convolutional networks to construct feature pyramids with marginal extra cost. A top-down architecture with lateral connections is developed for building high-level semantic feature maps at all scales. This architecture, called a Feature Pyramid Network (FPN), shows significant improvement as a generic feature extractor in several applications. He et al. (2019) proposed the performance of the road extraction network by integrating atrous spatial pyramid pooling (ASPP) with an Encoder-Decoder network. These methods effectively and reasonably enhance the receptive field, thereby improving the integration of road features. However, tree occlusion introduces noise into lower-level features, causing irregular road boundaries and disrupting skip connections between shallow and deep features. Consequently, it is unnecessary to employ skip connections at each level within the U-shaped structure.

The attention mechanism is pivotal in road extraction, enhancing feature representation and generally categorized into channel and spatial attention. Hu et al. (2018) proposed a novel architectural unit, which we term the Squeeze-and-Excitation (SE) block, that adaptively recalibrated channel-wise feature responses by explicitly modeling interdependencies between channels. The attention mechanism effectively captures global context and manages local feature details. However, it can introduce significant parameter complexity and computational load, consequently increasing time and memory costs during model training and inference. Wang et al. (2020) proposed an efficient Channel Attention (ECA) module that only involves a few parameters and brings significant performance improvements. By analyzing the channel attention module in SENet, appropriate cross channel interactions can significantly reduced model complexity while maintaining performance. While lightweight channel attention excels in amplifying features along the channel dimension, it often falls short in effectively

extracting features across the spatial dimension. To address this challenge, Zhuang et al. (2024) proposed the Channel-Spatial Attention Module (CSAM) by leveraging the power of channel attention and spatial attention. Zhou et al. (2021) proposed a position attention module and a channel attention module are added in parallel to capture the spatial dependency and the channel dependency, respectively. The two modules produce the fusion feature map to enhance the feature representation and improve network classification performance.

We introduced PCSAM in skip connections linking the final three layers of the encoder's spatial detail branch with the initial three layers of the decoder. In parallel, FFM was applied across all stages of the spatial detail branch, and strategically integrated into the skip connections between the global semantic branch and the decoder's terminal layer. Moreover, we employed ASPP to refine redundant information and expand the receptive field for the fused output features from the encoder's dual branches.

3. Proposed approach

3.1. Network architecture

Fig. 1 illustrates the overall architecture of LightCGS-Net, integrated the structural schematics of most semantic segmentation networks. The main framework structure of the network in this article consists of encoders and decoders. The encoder creates up a two-branch road feature fusion structure with simultaneously spatial detail branches and global semantic branches. The decoder adopts a usual convolutional upsampling processes to restore the image's dimensional size. Concurrently, FFM and PCSAM are intended to be added to the skip connection in order to further enhance the information extracted from HRSIs concerning boundary characteristics and occlusion.

The three distinct components that comprise the global semantic branch include patch embedding, ST block, and patch merging. It constitutes a feature downsampling structure based on ST. Simultaneously, we modify the overall structure to ensure that the model retains correctness with few parameters and processing. Based on the residual block, the spatial detail branch is a feature downsampling structure which includes a convolution operation for feature extraction and channel number mutation. So as to correspond with the processing speed capabilities of the model, we substitute the standard convolution in the original residual block with DW-Conv and PW-Conv for sparse decomposition parallel computation. The redundant information of the image filtered and sent to the decoder has been decreased using the Gaussian filtering noise reduction method. This lowers the quantities of overlap and redundant noise information generated by the overall global features and local detail features of the two branches of the element-wise addition fusion. Creating skip connections with D1, D3, D4, and D5 in the decoder between layer E1 in the global semantic branch and layers E2', E3', and E4' in the spatial detail branch. Additionally, a multi-scale feature fusion module referred to FFM is implemented in the skip connection between layer E1 and layer D1, introducing a cavity convolution. Increase the image's capability to extract edge information. To deal with the road incoherence features formed by the occlusion problem, a feature extraction module PCSAM based on lightweight attention strategy has been embedded in the remaining skip connections.

3.2. Global semantic branch based on light swin transformer

As shown in the E1-E5 downsampling stages in Fig. 1, the global semantic branch we created is based on the ST with lightweight fine-tuning. The branch can be divided into roughly four phases, with each phase consisting of the four parts that follow: Liner Embedding, Patch Partition, Patch Embedding, and ST Block, which each are followed by a patch merging procedure. At every phase, it is a multifaceted

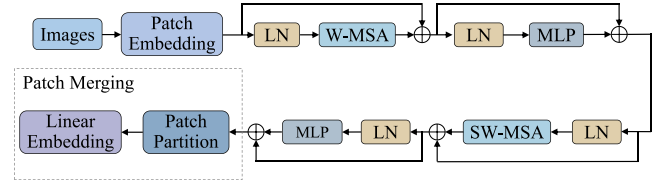


Fig. 2. Lightweight design of a Light Swin Transformer structure with a ST block that alternates among the application of W-MSA and SW-MSA. LN represents LayerNorm. The subsequent layer designates new W-MSA calculation, spanning across the boundaries of previous W-MSA in the preceding layer to provide connections across W-MSA. This operation is streamlined, transforming the module into LightST.

reduction in feature size and a two-fold rise in channel density. Upon extraction, the size as $2C \times H/2 \times W/2$.

With the original, high-resolution 1024×1024 image, we initially perform Patch Embedding. Then, we input the features that are generated into the ST Block. Following that, the chunks are processed by the Patch Partition module to finish the Patch Merging process. This results in output features with a 512×512 resolution and are transmitted into the second, third, and fourth extraction branch phases. Patch Embedding and numerous ST blocks are incorporated in these steps to create image features with 256×256 , 128×128 , and 64×64 resolutions, respectively. For the purpose of to define the remote feature interactions and gather global information, these four stages create a hierarchical structure that gradually reduces the feature resolution layer by layer and increases the receiver field.

The multi-head self-attention module, which adaptively learns the relationship between diverse input regions, is the fundamental component of ST block. The patch merge layer improves the resolution of the input feature mapping for downsampling, and serves primarily to compute self-attention. In particular, the whole image has been split up into various non-overlapping windows to facilitate self-attention computation, as well as each window contains 64 patches, each size 4×4 for self-attention. The image's blocks supply several stacked light transformer blocks via their inputs. Relative position information has been used for determining self-attention within a limited window:

$$Q = XW_Q \quad K = XW_K \quad V = XW_V \quad (1)$$

$$Attention(Q, K, V) = Softmax\left(B + \frac{QK^T}{\sqrt{d}}\right)V \quad (2)$$

where the dimension that defines the feature is d . Multiplying X by the parameter matrices W_Q , W_K , and W_V yields the query matrix Q , the key matrix K , and the value matrix V . Upon extracting the normalized weights using Q and K , the properties of interest are obtained. The relative position of the patches within a single window can be seen by B , which is the relative positional bias.

As shown in Fig. 2, The cross-window fusion of patch features can be achieved applying the window division of ST block and the shifted window scheme, which is compared to the classic ViT, significantly decreases the computing cost of the model. To permit the self-concerned computation in the new window divided in the later layer of the network to cross the boundaries of the windows in the original layer of the network, the Patch Embedding operation will be carried out initialized followed by the local Window MSA (W-MSA) and the Shifted Window MSA (SW-MSA) in alternative. It develops a communication connection across the local windows to capture the global contextual information. By employing the E3 layer as an example, the procedure is continued. For the purpose of to further broaden the reception field, adjacent patches are combined as well utilizing the patch merging layer. Before each MSA module, a LayerNorm gets used to normalize the features while enhancing the stability of the module structure. Simultaneously, each MSA module has a Multi-layer Perceptron (MLP) placed behind it so that the model is more capable of understanding the interactions

between features at multiple locations in the image. Finally, Patch Merging strengthens the perception of the spatial structure. The precise approach of computation is as follows:

$$\tilde{Z}^W = W_MSA - MSA(LayerNorm(\tilde{Z}^{W+1})) + Z^{W+1} \quad (3)$$

$$Z^M = MLP(LayerNorm(\tilde{Z}^W)) + \tilde{Z}^W \quad (4)$$

$$\tilde{Z}^{SW+1} = SW_MSA(LayerNorm(\tilde{Z}^M)) + Z^M \quad (5)$$

$$Z^{M+1} = MLP(LayerNorm(\tilde{Z}^{SW+1})) + \tilde{Z}^{SW+1} \quad (6)$$

where Z^{W+1} corresponds to the self-attentive estimation in the latter layer delimiting the new window, which crosses the boundary of the prior window in the layer preceding it and provides the connection across the local window, whereas $\tilde{Z}^W$ is the output feature of the W-MSA module. The outcome of the SW-MSA module's feature is $\tilde{Z}^{SW+1}$, the output of the first MLP module's feature is Z^M , and the output of the second MLP module's feature is Z^{M+1} .

3.3. Spatial detail branch based on lightCNN

We propose a lightweight CNN-based detail feature extraction branch with fewer parameters, where the convolution in the residual block is decomposed into DW-Conv for spatial feature extraction and PW-Conv for channel feature extraction. This forms the lightweight downsampling convolution layer in our improved residual block. By applying a unified, lightweight design for the residual sub-module, we address the redundancy in conventional network structures, thereby reducing both computational load and parameter count, while enhancing image processing time and inference speed.

The conversion convolution process is carried out using a 3×3 DW-Conv with a step size of 2, and the number of channels is then converted using PW-Conv. With the first convolution operation, we add a secondary sampling layer using two sets of 2×2 convolutional conversion layers with a step size of 2, which makes it possible to increase by double the size of the channel information and halve the aspect information of the image size. To stabilize the training, activation function layers and batch normalization have been set up simultaneously. The second 3×3 DW-Conv is subsequently separated into 1×3 DW-Conv and 3×1 DW-Conv layer sparse decomposition for parallel computation. PW-Conv is used for sub-module downsampling and channel superposition, and low-rank branching is designed in parallel to decrease computational cost. Moreover, we adopt the lightweight activation function Switch with the conventional ReLU function. This function performs noticeably better than the ReLU function in this model, especially shown through the experimental comparison. In the interim, during model extraction, the lightweight design minimizes the amount of complicated memory and processing power consumed by the occupied device.

3.4. Parallel channel space attention mechanism

Channel Attention Mechanism (CAM) combines average pooling kernels $H \times 1$ and $1 \times W$ with differently interpreted channel information to carry out the average pooling functioning at input feature mapping. The dimensions are then transformed into two feature tensors, $f_{Horizontal} \in R^{C \times H \times 1}$ and $f_{vertical} \in R^{C \times 1 \times W}$, after which the associated dimensions are combined into $C \times 1 \times 1$, using the Sigmoid activation function to obtain the channel weights of the vertical and horizontal features, respectively. For generating a new feature tensor $f_{CAM} \in R^{2C \times H \times W}$, the feature map is concatenated. 1×1 Conv is the method employed to shrink the channel, and $C \times H \times W$ is the final output dimension. The precise approach for computation is as follows:

$$X_{Horizontal} = Sigmoid(Conv(f_{Horizontal})) \quad (7)$$

$$X_{vertical} = Sigmoid(Conv(f_{vertical})) \quad (8)$$

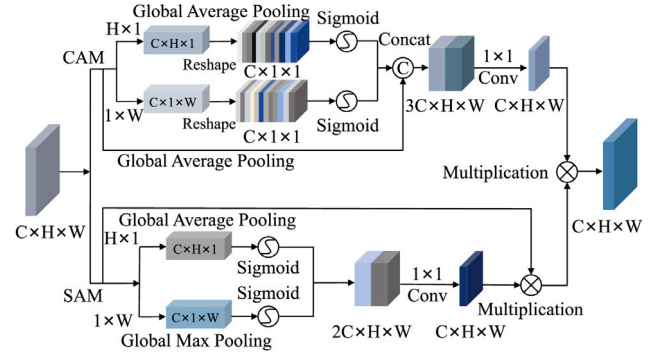


Fig. 3. Structure of PCSAM Module. The upper section represents the CAM, which emphasizes the interrelationships and dependencies between channels. The lower section represents the SAM, which focuses on feature representation across different spatial locations.

$$F_{CAM} = Concatenate(X_{Horizontal}, X_{vertical}) \quad (9)$$

During the input feature mapping of $C \times H \times W$, the Spatial Attention Mechanism (SAM) splits the two-dimensional global average pooling operation into two one-dimensional global average pooling operations. Using an average pooling kernel of size $(H, 1)$, the Global Average Pooling (GAP) compression operation is done on the features along the horizontal direction. Using a maximum pooling kernel of kernel size $(1, W)$. The Global Max Pooling (GMP) compression operation is performed to the features along the vertical direction. The dimensions are transformed into a pair of feature tensors, $f_{Horizontal} \in R^{C \times H \times 1}$ and $f_{vertical} \in R^{C \times 1 \times W}$, respectively containing distinct spatial information. This prompts the attention module to capture accurate location information along with remote spatial interactions. The new feature tensor can be obtained through the use of the Sigmoid activation function to output the position weights of the feature mapping after the position weight-weighted feature map has been combined. The channel has been reduced via the 1×1 Conv, and the final output dimension is $C \times H \times W$. Below is the exact calculating process:

$$Y_{Horizontal} = (Conv(f_{Horizontal})) \quad (10)$$

$$Y_{vertical} = (Conv(f_{vertical})) \quad (11)$$

$$F_{SAM} = Sigmoid(Concatenate(Y_{Horizontal}, Y_{vertical})) \quad (12)$$

The PCSAM module is a matrix multiplication operation that uses the light-weight activation function Switch to normalize the channel attention-enhanced feature F_{CAM} and the spatial attention-enhanced feature F_{SAM} following overlaying their tensors and establishing them by means of BN layers. The exact process of calculation is as follows:

$$Z_{PCSAM} = Switch(BN(Mul(F_{CAM}, F_{SAM}))) \quad (13)$$

The complex layout of the PCSAM is illustrated in Fig. 3, where the SAM is positioned at the bottom and the CAM at the top. This design not only considers the dependencies between channels but also addresses the representation of features across various spatial locations. By focusing on both horizontal and vertical directions of the inputs, each tensor element facilitates attention optimization across multiple dimensions, enabling the model to accurately identify road areas within each feature channel. Additionally, connecting the two attention modules in parallel with a lightweight activation function simplifies and enhances the efficiency of capturing weighting information in both the channel and spatial dimensions. Furthermore, our Ablation Study demonstrates that PCSAM outperforms traditional attention mechanisms in evaluation metrics, indicating that the proposed module is more effective for segmenting roadway characteristics in HRSIs.

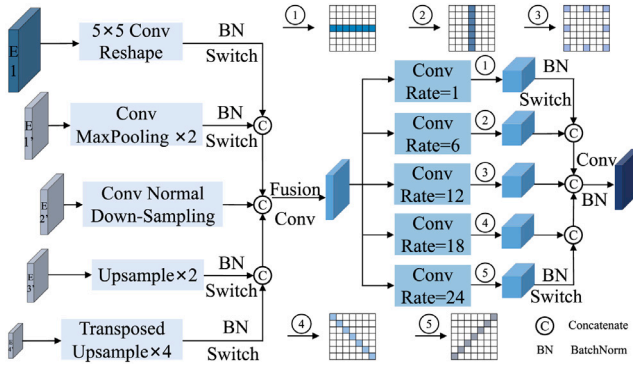


Fig. 4. Structure of FFM. It captures multi-level information from different stages of the encoder. The deep network extracts features that provide road semantic information, enabling the distinction of roads from the background. The shallow network extracts features rich in spatial information, allowing for precise road localization.

3.5. Feature fusion mechanism

In order to fuse the features at different scales in each stage of spatial detail branch in the encoder, we created the Feature Fusion Module (FFM). In Fig. 4, the FFM is exhibited. Multi-scale features are fused with the E1 layer in the global feature branch and the E1', E2', E3', and E4' layers in the spatial detail branch. The fused features are subsequently spliced using the stitching procedure. E4' layers for multi-scale feature fusion. The fused features are subsequently utilized for splicing, and the spliced features are subsequently submitted to ASPP for sensory computation based on their convolution layers that have various unfolding rates to obtain the global and edge features of multiple images. The aforementioned procedure is carried out for vertical, horizontal, oblique, and edge field of view. These images are initially up-sampled via transposed convolution to render them the same size and proportions as the final layer of the decoder, then they are subjected to an additional stitching handle which enhances the model's capacity to extract edge features. For altering the channel as the final output of the image at the decoder side, channel cascading with 1×1 Conv eventually gets sent out.

The E2' layer in the spatial detail branch provides the foundation for this specific principle. The fine-grained feature maps extracted from the E3' and E4' layers of the latter downsampling are subsequently put through to the inverse convolution operation with four 2×2 kernels and the linear interpolation method, respectively, to expand the feature sizes of each layer uniformly to the same size as that of the E2' layer features. Following getting coarse-grained feature maps from the E1 layer in the global feature branch and the E1' layer in the spatial detail branch, the feature maps experienced 5×5 convolutional transformation and pooling layer downsampling strategies to correspond with the output size of the E2' layer. Lightweight grouped convolution was employed to decrease the number of channels to 64, and the feature maps seemed spliced. Applying different expansion rate operations in the airspace convolution, the spliced feature maps have been used to extract global road information and edge road information with ASPP. In order to generate accurate road structure feature maps for the final FFM output, the improved data is placed via a secondary splicing operation. The follows by BN and ReLU combined with the additional road structure information. Subsequently, the transposed convolution upsamples the FFM-fused feature images to ensure that they have the same dimensional size as the last layer of D1 on the decoder side. They perform channel cascade operation, and 1×1 convolution is employed to alter the number of channels. The decoder output image's edge information is deteriorated, which significantly improves the model's capacity to identify and split off the road features in HRSIs.

4. Experiments and results

4.1. Experimental road datasets

As a way to assess the efficacy of the proposed approach, we use three publicly accessible road datasets (DeepGlobe (Demir et al., 2018), Massachusetts (Forman and Deblinger, 2000), and SpaceNet (Van Etten et al., 2018)) as the experimental datasets for semantic segmentation network modeling. These datasets enable the extraction of road information from remotely sensed images. They are publicly accessible and frequently used in research on semantic segmentation methods for traffic data.

The DeepGlobe road datasets provide road information across diverse scenes and environments. The satellite images, captured by WorldView-2/3, are in RGB format with a ground resolution of approximately 50 cm per pixel. This dataset includes 6226 pairs of 1024×1024 pixel images from urban and rural areas in India, Thailand, and Indonesia. Due to the lack of label annotations in the validation and test sets, we divided the dataset into 4359 training sets, 1245 validation sets, and 622 test sets in a 7:2:1 ratio for our experiments.

The Massachusetts road datasets consist of 1171 RGB aerial images, each measuring 1500×1500 pixels, encompassing urban, suburban, and rural areas. Each image has a resolution of 1.0 m per pixel, covering approximately 2600 square kilometers of landscape in Massachusetts. Each image is accompanied by a grayscale mask, where white pixels represent roads and black pixels indicate the background. Within the dataset, 1108 images are designated for training, 49 for testing, and 14 for validation.

The SpaceNet road datasets is a collection of HRSIs, each measuring 1300×1300 pixels, made available through the SpaceNet Challenge 2018. This dataset includes corresponding road outlines from various regions worldwide, such as Khartoum, Paris, Shanghai, and Las Vegas. The road network data is meticulously hand-labeled to ensure accuracy and quality. For our experiments, we selected SpaceNet 03–05 as the training, validation, and test sets, encompassing the regions of Mumbai, Paris, Shanghai, Khartoum, and Las Vegas (see Fig. 5).

4.2. Configurations and implementation details

We employed the PyTorch deep learning framework, utilizing two NVIDIA RTX 4090 GPUs for model training and testing. The complexity of the network was assessed by measuring the number of parameters and flops (Kollár, 1989). Prior to training, the training images were augmented through horizontal or vertical flipping, grid warping, and 90° rotation methods, resulting in a threefold increase in the number of training images compared to the original dataset.

The network accepts input images of 1500×1500 pixels from the Massachusetts dataset. Padding is applied to resize the image to 1504×1504 pixels, followed by random cropping to obtain the original size of 1024×1024 pixels. The images from the DeepGlobe dataset are grayscaled and also sized at 1024×1024 pixels, matching the network input. Images from the SpaceNet dataset, originally 1300×1300 pixels, are reduced to 1024×1024 pixels. This preprocessing ensures that all three datasets are unified and compatible with the model's network architecture prior to input.

The training batch size is set to 4, with an initial learning rate of 0.0001, and a total of 200 epochs for training and validation. The Adam optimizer is employed to minimize the loss function and identify optimal control parameters. A custom learning rate decay technique is implemented, which saves the model's weights and updates training losses based on the assumption that the loss values in the new round converge with those of the previous round. If the training loss does not improve for three consecutive epochs, the learning rate is decreased by a factor of 5 for the next training session.



Fig. 5. Three road dataset image examples. The first row is DeepGlobe road datasets, the second row is Massachusetts road datasets, and the third row is SpaceNet road datasets.

4.3. Hybrid loss

This paper presents a hybrid loss function (BCE-Dice) that combines the Binary Cross-Entropy (BCE) (Ruby and Yendapalli, 2020) loss and Dice Coefficient (Dice) loss (Sudre et al., 2017) to enhance model segmentation performance. BCE loss captures structural information to preserve image texture, while Dice loss emphasizes edge details, particularly in high-frequency regions. This integration ensures the extraction of fine edge features while maintaining global feature integrity. The specific formulation of the loss function is as follows:

$$L_{BCE} = 1 - \frac{2 \sum_i^{H \times W} \varphi_{true}^i \varphi_{pred}^i + 1}{\sum_i^{H \times W} (\varphi_{true}^i)^2 + \sum_i^{H \times W} (\varphi_{pred}^i)^2 + 1} \quad (14)$$

$$L_{Dice} = -\frac{1}{N} \sum_{i=1}^N \left(\gamma \varphi_{true}^i \log(\varphi_{pred}^i) + (1 - \varphi_{true}^i) \log(1 - \varphi_{pred}^i) \right) \quad (15)$$

$$L_{Hybrid} = L_{BCE}(\delta_{true}, \delta_{pred}) + L_{Dice}(\delta_{true}, \delta_{pred}) \quad (16)$$

The number of pixels which utilize the sample is N . Pixel i 's binary class label, indicated as φ_{true}^i , represents the true label of the sample, with a value of either 0 or 1. The anticipated probability of the sample categorization, indicated by φ_{pred}^i , ranges from 0 to 1. The segmentation models and edge information extraction are rendered more accurate to varying degrees by the loss functions L_{Road} and L_{Edge} for the edge details and for the sample label, for the prediction result, δ_{true} and δ_{pred} for the overall road texture.

4.4. Evaluation metrics

We evaluate the model's extraction effectiveness by five evaluation metrics: Precision, Recall, IoU, F1-score, and Accuracy. Specifically, Precision represents the ratio of correctly predicted road pixels to all pixels predicted as roads, while Recall measures the ratio of correctly predicted road pixels to all actual road pixels in the ground truth image. F1-score is the harmonic mean of Precision and Recall. IoU reflects the proportion of overlap between predicted road pixels and ground

truth pixels. Accuracy refers to the percentage of correct predictions relative to total predictions. Higher values of these metrics indicate better performance of the road extraction model. The specific metric definitions are as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (17)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (18)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + FN + FP + TN} \quad (19)$$

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP + FN + FP} \quad (20)$$

$$\text{IoU} = \frac{TP}{TP + TN + FP} \quad (21)$$

where TP and FP represent to true positives and false positives, respectively, in the road prediction results, which refers to the total number of road background pixels and the number of correct road pixels. FP and FN reflect the number of true-negative and false-negative elements in the correctly and incorrectly predicted negative category results, respectively, whereas FN denotes the number of road background pixels. The adequately identified background pixels are conveyed as TN.

4.5. Evaluation of model performance

As our comparison networks for the experiment, we choose U-Net, D-LinkNet, NL-LinkNet, RoadFormer, LG-Net, SwinUNet, and TransUNet, TransRoadNet. We contrast our LightCGS-Net's segmentation accuracy and performance to currently complex road extraction networks with three public road datasets (DeepGlobe, Massachusetts, and SpaceNet) and assess its efficacy in terms of Precision, Recall, IoU, F1-score, and Accuracy. As an example of the superior segmentation accuracy and performance of our LightCGS-Net construction over battling innovative road extraction networks.

The quantitative results for multiple classical techniques are shown in Table 1, that allows for the conclusion that our proposed LightCGS-Net obtains the best IoU metrics (0.7123, 0.6857, and 0.7055) on the three datasets, regardless of sequence. On DeepGlobe and Massachusetts, the best F1-score (0.8146 and 0.8087) and Accuracy (0.9895 and 0.9872) were obtained. In contrast with D-LinkNet, LightCGS-Net's

Table 1

Compares the road feature extraction performance of different segmentation models under the three public road data sets of DeepGlobe, Massachusetts and SpaceNet.

Model	Datasets	Precision	IoU	Recall	F1-score	Accuracy
U-Net	DeepGlobe	0.7561	0.6193	0.8008	0.7544	0.9687
	SpaceNet	0.7396	0.5802	0.7973	0.7721	0.9704
	Massachusetts	0.7976	0.6269	0.7412	0.7649	0.9711
D-LinkNet	DeepGlobe	0.7531	0.6322	0.8036	0.7631	0.9822
	SpaceNet	0.7691	0.6441	0.8111	0.7559	0.9841
	Massachusetts	0.7896	0.6287	0.7496	0.7724	0.9838
NL-LinkNet	DeepGlobe	0.7867	0.6579	0.8076	0.7783	0.9788
	SpaceNet	0.7384	0.6333	0.7717	0.7894	0.9756
	Massachusetts	0.7367	0.6247	0.7697	0.7882	0.9782
SwinUNet	DeepGlobe	0.8201	0.6227	0.7206	0.7671	0.9813
	SpaceNet	0.7896	0.6401	0.7366	0.7811	0.9823
	Massachusetts	0.7922	0.6216	0.7145	0.7513	0.9846
TransUNet	DeepGlobe	0.8033	0.6239	0.7364	0.7684	0.9821
	SpaceNet	0.8034	0.6723	0.8107	0.7913	0.9847
	Massachusetts	0.7711	0.6082	0.7125	0.7406	0.9799
LG-Net	DeepGlobe	0.7996	0.6829	0.8269	0.8346	0.9864
	SpaceNet	0.8127	0.6436	0.7574	0.7801	0.9877
	Massachusetts	0.7673	0.6441	0.7845	0.7898	0.9858
RoadFormer	DeepGlobe	0.8227	0.6722	0.7819	0.7915	0.9773
	SpaceNet	0.8354	0.6704	0.8017	0.8205	0.9786
	Massachusetts	0.8054	0.6646	0.7894	0.7971	0.9778
TransRoadNet	DeepGlobe	0.8101	0.7021	0.8413	0.8134	0.9866
	SpaceNet	0.8108	0.7176	0.8346	0.7994	0.9845
	Massachusetts	0.8066	0.7041	0.8221	0.7993	0.9817
LightCGS-Net	DeepGlobe	0.8029	0.7123	0.8391	0.8146	0.9895
	SpaceNet	0.8144	0.6857	0.8374	0.8018	0.9841
	Massachusetts	0.8033	0.7055	0.8246	0.8087	0.9872

recall improves by 2.8%, 2.63%, and 2.96%, respectively, indicating that our model effectively extracts the entire road network structure. Our models all achieve superior metrics when compared to traditional road network (NL-LinkNet), suggesting that the road labels that LightCGS-Net obtained are more compatible with truth. Our network surpassed LG-Net in each additional a parameter, notwithstanding their superior results in Recall and F1-score, respectively. LightCGS-Net recognizes a perfect framework that surpassed the road segmentation tackles that integrate Transformer (SwinUNet, TransUNet, RoadFormer, TransRoadNet) in terms of overall performance, although proving worse to them in one metric on all three datasets. It is shown that the method we recommend surpasses other models in the identification of pixel categories. Likewise LightCGS-Net accomplishes excellent performance with fewer computational complexity and parameters compared to other Transformer-based road segmentation networks, as well as fewer flops.

We deduced from the analysis that the reason why the Precision metrics of LightCGS-Net (0.8029, 0.8144, and 0.8033) on all three datasets dropped brief of the optimal metrics in the comparison network of this dataset was due to some areas of the dataset possessed incomplete images and background gaps, leading to some false areas' feature fine information to have been incorrectly categorized. However, it seems apparent that LightCGS-Net performed better overall than the other segmentation models. It reveals the method's excellent generalization and domination in completing the task of eliminating roadways from HRSIs.

4.6. Ablation study

To validate the contributions of each module in our propose LightCGS-Net, we conducted ablation experiments using LinkNet34 (Chaurasia and Culurciello, 2017) as the baseline model. The model's performance metrics improved significantly after integrating the Transformer, and the optimal results were achieved after introducing the PCSAM and FFM modules, demonstrating the necessity and effectiveness of our module designs. Additionally, we visualized and analyzed the results of the ablation experiments through feature heatmaps for

comparative analysis. Furthermore, we performed experimental analyses on the selection of the loss function and the ratio of the two components in the composite loss, identifying the most suitable loss configuration for our network in terms of convergence speed and loss performance.

(1) *Effectiveness Analysis of LightCGS-Net*: Table 2 demonstrates that the network consistently outperforms the Baseline on both the DeepGlobe and Massachusetts datasets across all evaluation metrics, attributed to the decoder's ST-based global semantic branching. This approach enhances global feature extraction by utilizing self-attention for capturing long-range dependencies. The segmentation accuracy of global roads and edge details is significantly improved through multi-scale feature fusion and null convolution, which increases the convolutional kernel's receptive field. The Recall decreases from 83.91% and 82.46% to 83.76% and 81.44% when FFM is removed, confirming the effectiveness of our dual-attention structure in aggregating road features and addressing road discontinuities due to occlusions, even without PCSAM. Precision decreases by 1.63% and 0.95%, while IoU drops by 0.82% and 1.55%. On the SpaceNet dataset, the designed modules also outperform the Baseline, demonstrating their effectiveness and synergy in forming a robust LightCGS-Net. The introduction of ST significantly improves the LinkNet34 network across multiple metrics. PCSAM enhances feature extraction via its attention mechanism, achieving an optimal F1-score, although other metrics decline. FFM further improves recall and F1-score, with the final network reaching optimal IoU and recall, though other metrics remain suboptimal. This is likely due to SpaceNet's low pixel density, which leads to false background segmentation and reduced precision in extracting detailed road information.

(2) *Visual feature heatmap comparison*: Fig. 6 indicates the feature map comparison of the last layer, which is visualized as a feature heatmap adhering to the introduction of each module in the ablation experiment. The input and label corresponds of the HRSIs are shown for (a) and (c), the heatmap based on background information is for (b), the segmentation heatmap of LinkNet34 is by (d), and the heatmaps after of introducing Transformer and PCSAM are for (e) and (f), respectively. The network clearly learns more plentiful background

Table 2

The evaluation metrics of the innovative modules in LightCGS-Net across three datasets.

Components	DeepGlobe				Massachusetts				SpaceNet			
	Precision	IoU	Recall	F1	Precision	IoU	Recall	F1	Precision	IoU	Recall	F1
Baseline (LinkNet34)	0.7467	0.6768	0.8107	0.7913	0.7749	0.6587	0.8297	0.7834	0.7635	0.6312	0.8157	0.7439
Baseline + ST	0.7967	0.6897	0.8254	0.8017	0.8161	0.6741	0.8311	0.7922	0.7933	0.6447	0.7736	0.7804
Baseline + ST + PCSAM	0.8134	0.7109	0.8376	0.8029	0.8066	0.6699	0.8144	0.7881	0.7577	0.6273	0.7941	0.8047
Baseline + ST + FFM	0.7866	0.7041	0.8401	0.8111	0.8049	0.6702	0.8271	0.7917	0.7879	0.6385	0.8218	0.7897
LightCGS-Net	0.8029	0.7123	0.8391	0.8146	0.8033	0.7055	0.8246	0.8087	0.7912	0.6527	0.8234	0.7906

Table 3

The comparative results of LightCGS-Net, which incorporates various attention modules in the skip connection, on the DeepGlobe, Massachusetts, and SpaceNet road datasets.

Skip Connection	DeepGlobe				Massachusetts				SpaceNet 03-05			
	Precision	IoU	Recall	F1	Precision	IoU	Recall	F1	Precision	IoU	Recall	F1
Without PCSAM	0.7723	0.6902	0.7135	0.8064	0.7615	0.6567	0.7814	0.7711	0.7687	0.6567	0.7927	0.7581
with SE	0.7914	0.6681	0.7371	0.7898	0.7397	0.6396	0.7992	0.7690	0.7712	0.6396	0.7663	0.7661
with ECA	0.7898	0.6866	0.7299	0.8106	0.7522	0.6534	0.8007	0.7856	0.7766	0.6534	0.8134	0.7575
with CSAM	0.7791	0.6818	0.7154	0.8012	0.7467	0.6471	0.8311	0.7764	0.7884	0.6471	0.8178	0.7901
with CPAM	0.8139	0.6943	0.7216	0.8101	0.7789	0.6602	0.8266	0.7956	0.7947	0.6602	0.7934	0.7769
with PCSAM	0.8029	0.7123	0.8391	0.8146	0.8033	0.7055	0.8246	0.8087	0.7912	0.6527	0.8234	0.7906

Table 4

Comparison of IoU/F1 as road segmentation results on three datasets using LightCGS-Net with different loss functions.

Loss Function	Dataset		
	DeepGlobe	Massachusetts	SpaceNet
BCE	67.74/79.64	67.11/75.67	60.07/77.14
Dice	66.37/77.41	67.23/77.67	60.23/79.04
WCE	68.74/79.84	68.77/77.36	62.79/78.74
Jaccard	68.21/77.41	68.37/76.41	61.11/78.67
Focal	70.27/77.31	69.11/77.74	61.74/80.57
Tversky	68.77/78.45	68.61/78.96	63.67/79.78
Combo	69.94/80.03	71.11/78.47	62.32/81.49
WCE-Dice	70.27/79.31	69.11/79.55	63.01/80.66
BCE-Dice	71.23/81.46	70.55/80.87	65.27/82.34

Table 5

Comparison of road segmentation results on three datasets using LightCGS-Net with different proportion of loss functions.

BCE : Dice	Precision (%)	Recall (%)	IoU (%)	F1-score (%)
Results of road segmentation on DeepGlobe datasets				
1 : 1	71.64	82.07	69.11	80.17
1 : 2	70.29	83.91	71.23	81.46
1 : 3	70.45	81.14	66.67	79.28
2 : 1	70.38	82.61	70.11	77.67
3 : 1	70.17	82.37	65.71	73.09
Results of road segmentation on Massachusetts datasets				
1 : 1	71.47	81.06	67.11	77.97
1 : 2	80.33	82.46	70.55	80.87
1 : 3	69.68	79.77	66.14	78.46
2 : 1	69.92	79.14	66.51	79.01
3 : 1	71.03	79.88	66.97	77.74
Results of road segmentation on SpaceNet datasets				
1 : 1	76.54	81.99	65.16	80.23
1 : 2	77.21	82.34	65.27	82.34
1 : 3	75.69	79.97	63.23	78.14
2 : 1	77.36	78.76	61.59	77.12
3 : 1	72.33	74.36	59.37	74.84

features with greater detail, enabling more comprehensive semantic representations. PCSAM aids in extracting roads in complex, occluded environments, offering more accurate and continuous detection, particularly for smaller roads. (g) presents our final network model. The figure demonstrates how adding FFM enhances the model's ability to extract edge road features while reducing redundancy, highlighting its capacity to incorporate global context and eliminate unnecessary information.

(3) *Comparative analysis of attention mechanism:* Table 3 presents a performance comparison of various classic attention modules replacing our PCSAM. SE and ECA are channel attention modules, while CBAM and CPAM can be considered sequentially connected channel and spatial attention modules. The network's Recall and F1-score improved by 1%–2% across all three datasets after incorporating the attention modules. Compared to SE and ECA, although SE achieved the best Recall on DeepGlobe, its overall performance was inferior to our network in other aspects. In comparison to CSAM and CPAM, PCSAM achieved the best Recall and IoU on DeepGlobe and Massachusetts datasets. On the SpaceNet dataset, it obtained the highest Precision, IoU, and F1-score. Even though our performance in certain aspects was slightly lower, overall, the PCSAM we designed and selected outperformed on all three datasets. This not only demonstrates that PCSAM enables effective feature extraction but also confirms the generalization capability of our module design.

(4) *Selection of loss function:* Table 4 summarizes the comparison of the performance metrics (IoU/F1) obtained by LightCGS-Net using

different loss functions across three datasets, demonstrating that the hybrid loss we selected is the most optimal. Overall, the hybrid loss outperforms the single-loss in terms of metrics across all three datasets. This is because the hybrid loss leverages the complementary nature of region-based and distribution-based losses, allowing it to retain the overall image features while extracting finer details. Specifically, both BCE and Dice generally yield positive performance metrics. Focal loss (Mukhoti et al., 2020) results fall below average, leading to worse accuracy compared to similar approaches. As an improved version of BCE loss, WCE loss (Laiz et al., 2019) clearly performs better than BCE across the three datasets, as shown in the table, but its performance declines when combined with Dice in the hybrid loss. Jaccard Coefficient loss (Niwattanakul et al., 2013) and Tversky loss (Salehi et al., 2017) also maintain competitiveness, but they are not our best choice. Combo loss (Taghanaki et al., 2019) achieves the highest IoU value on the Massachusetts dataset due to the complementary benefits of its additional loss, but it shows no significant advantage in other datasets. BCE-Dice holds a slight edge in model performance over Combo and WCE-Dice losses across the three datasets and demonstrates

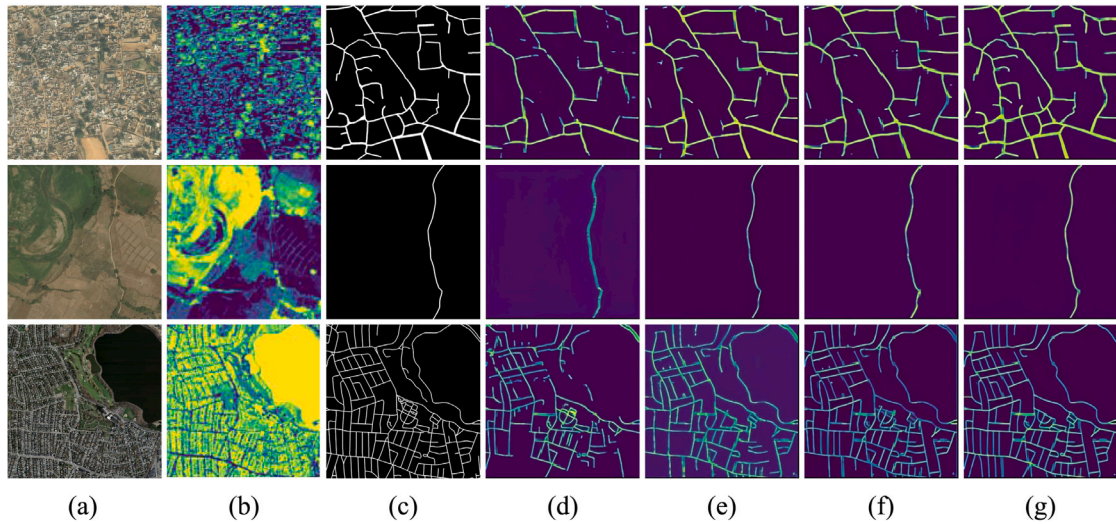


Fig. 6. Visualization of the feature maps of the last decoder. (a) Input images, (b) Background features, (c) Ground truths, (d) Baseline, (e) add ST, (f) w/o FFM, (g) LightCGS-Net.

strong superiority in the comparison of single losses. Therefore, after comparing various loss functions, we ultimately selected BCE-Dice as our loss function.

(5) *Analysis of the proportional impact of hybrid loss:* Table 5 presents a comparison of various performance metrics for hybrid loss (BCE-Dice) across three datasets under different weight ratios. We found that the 1:2 weighting ratio yielded the best results, which we adopted as the final loss ratio. Specifically, the 1:1 ratio maintains good overall model performance—although not optimal, it closely approximates the performance of the 1:2 ratio. Across all three datasets, the model performs suboptimally when the BCE weight exceeds that of Dice, while higher Dice weights generally lead to better outcomes. The performance is particularly strong in the DeepGlobe and Massachusetts datasets, with three metrics achieving optimal results, and two metrics reaching optimal levels in the SpaceNet dataset. Even for metrics that did not reach optimal values, our results remain close to the best-performing metrics. We also conducted ratio experiments for WCE-Dice, finding that the optimal ratio was 1:1, with the results recorded in Table 4. Additionally, we observed that larger ratios negatively impact model performance. We speculate that this decline occurs because, with disproportionate weight distribution between region-based and distribution-based losses, the effect of the lesser-weighted loss diminishes or even disappears. This not only prevents the model from achieving the ideal segmentation results but also introduces unnecessary computation, ultimately increasing the computational burden and processing time.

4.7. Model parameter and inference time

Table 6 compares the LightCGS-Net with TransUNet and TransRoadNet follows the lightweight treatment in terms of 139.46G and 125.49G fewer flops, and 51.71M and 24.43M fewer parameters, respectively. While this continues to develop on, LightCGS-Net has less flop than SwinUNet while containing 12.239M more parameters after the PCSAM and FFM modules are added. Meanwhile, the number of parameters and flops in our network is less compared to that of other indicated networks based on or incorporated with Transformer (TransUNet, RoadFormer, TransRoadNet).

We evaluate the time necessary by various network models to process a single image and snap the comparisons using 4090 GPUs. In terms of overall processing time, we find that the traditional road segmentation network performs more effectively than the network which utilizes the Transformer. For the three datasets, LightCGS-Net trails below SwinUNet approximately 27%, 25%, and 26%, respectively. As we suggest in Evaluation of Model Performance (Section 4.4), where we

exhibit that our extraction accuracy improves substantially compared to SwinUNet, we analyze the reason for this: SwinUNet uses U-Net as the backbone network, whereas our design is a parallel extraction network with CNN and Transformer in the encoder. This leads to the computational complexity over this network. Additionally, we surpass other Transformer road extraction networks in terms of both speed and inference time. The previously mentioned evaluations demonstrate the efficiency of the lightweight methodology. We have achieved the experimental objective of finding the optimal balance between segmentation accuracy and computational efficiency in LightCGS-Net.

4.8. Analysis of experimental visualization results

As shown in Fig. 7, the visualization results of eight segmentation networks on the DeepGlobe test set are presented. U-Net struggles to effectively handle situations where roads are occluded by trees or buildings, while D-LinkNet, through standard dilated convolutions, introduces contextual information that allows partially occluded roads to be extracted. Compared to Transformer-based road segmentation networks, the prediction results of LightCGS-Net are closer to the ground truth, with more complete segmentation details. This is because the global semantic branch integrates global contextual information of the roads, enhancing the long-distance dependencies between roads in HRSIs. Additionally, the dual attention mechanism design enables more accurate segmentation of small roads. The red bounding box specifically highlights the notable differences in segmentation between our network and various road networks, while the yellow bounding box demonstrates our method's superiority in extracting edge feature information compared to other models. It can be observed that, compared to traditional methods, LightCGS-Net accurately identifies and segments detailed features and edge information. The experimental results validate LightCGS-Net's ability to address the problem of road occlusion and provide more complete road extraction results in terms of image details and boundary information. Both the experimental and visualization results demonstrate that our network possesses the capability to segment fine features effectively.

Fig. 8 presents the prediction results of eight selected networks for segmenting fine features of large-scale objects on the Massachusetts dataset. The red and yellow bounding boxes have the same significance as in the DeepGlobe dataset. Compared to other road networks, our method captures significantly more detailed features in the surrounding areas. When processing complex real-world scenes with diverse backgrounds, LightCGS-Net accurately distinguishes small-scale objects from the background, effectively detecting detailed and

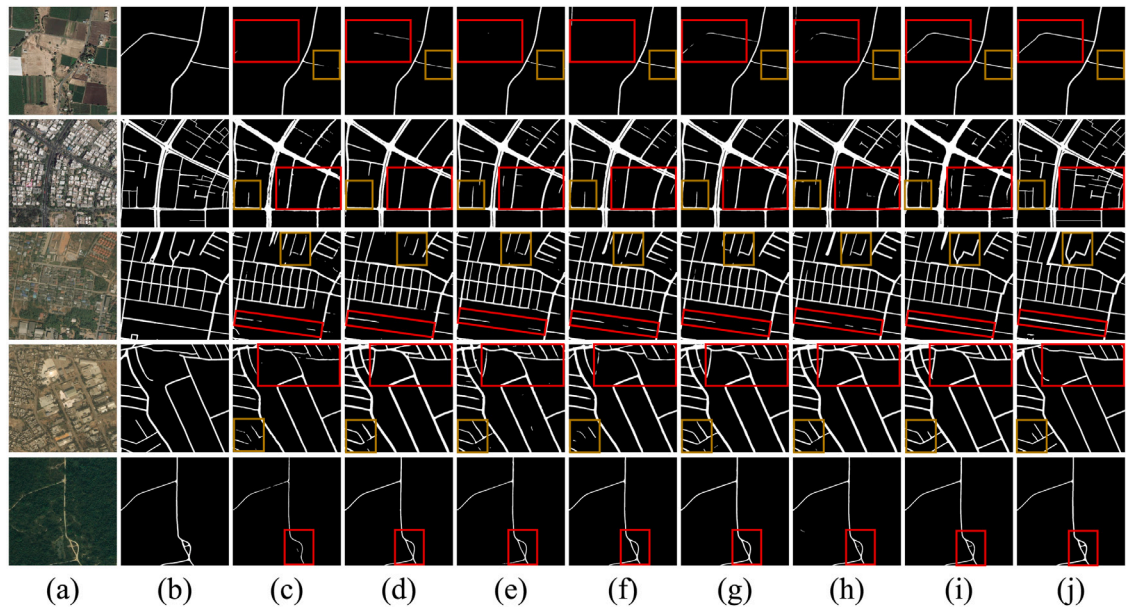


Fig. 7. Visualization results of different methods for DeepGlobe road datasets. (a) Input Image, (b) Ground Truth, (c) U-Net, (d) D-LinkNet, (e) LG-Net, (f) SwinUNet, (g) RoadFormer, (h) TransUNet, (i) TransRoadNet, (j) LightCGS-Net.

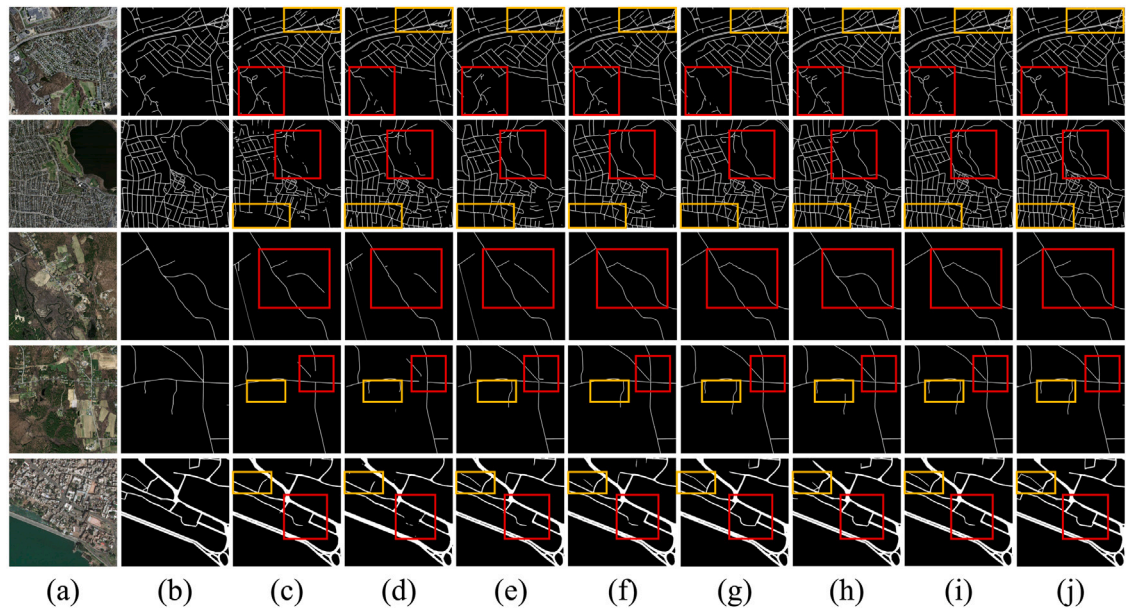


Fig. 8. Visualization results of Massachusetts road datasets using different methods. (a) Input Image, (b) Ground Truth, (c) U-Net, (d) D-LinkNet, (e) LG-Net, (f) SwinUNet, (g) RoadFormer, (h) TransUNet, (i) TransRoadNet, (j) LightCGS-Net.

Table 6

The comparison of various road segmentation models in terms of the time required to process a single HRSIs across three test sets using 4090 GPUs, along with the parameters and flops of each network model.

Model	Params.	FLOPs	DeepGlobe	Massachusetts	SpaceNet
U-Net	29.1M	35.15G	0.26s	0.24s	0.25s
D-LinkNet	31.08M	134.32G	0.36s	0.39s	0.37s
NL-LinkNet	21.64M	109.85G	0.33s	0.36s	0.34s
SwinUNet	27.1M	254.8G	0.35s	0.33s	0.34s
TransUNet	95.49M	335.55G	2.64s	2.58s	2.61s
LG-Net	27.66M	7.09G	0.26s	0.24s	0.28s
RoadFormer	48.02M	212.65G	0.63s	0.66s	0.63s
TransRoadNet	68.21M	321.58G	0.59s	0.59s	0.63s
LightCGS-Net	43.78M	196.09G	0.48s	0.44s	0.46s

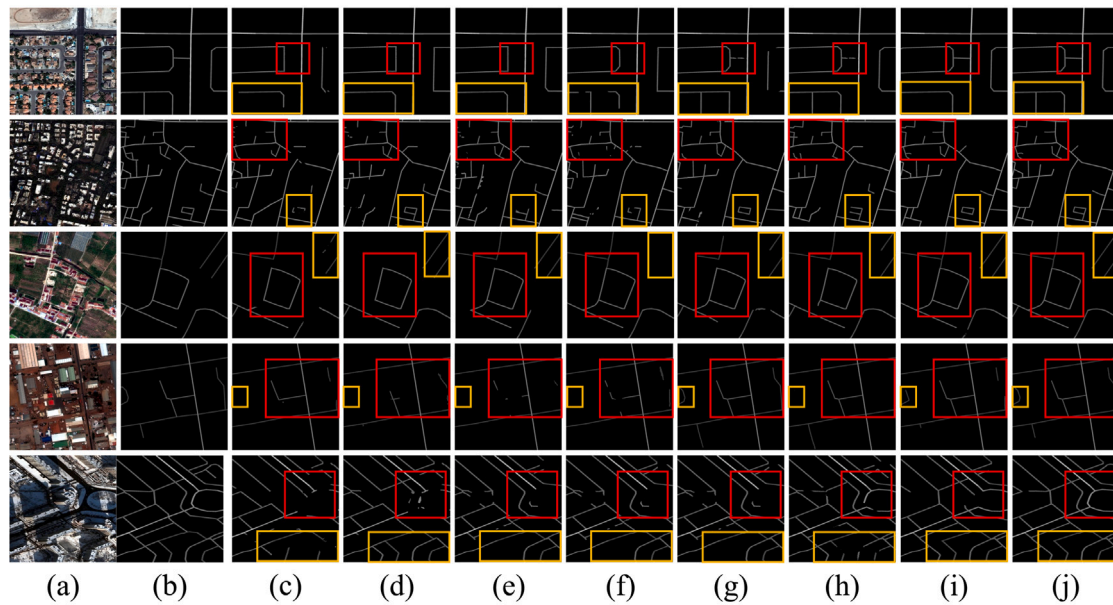


Fig. 9. Visualization results of SpaceNet 03-05 using different methods. (a) Input Image, (b) Ground Truth, (c) U-Net, (d) D-LinkNet, (e) LG-Net, (f) SwinUNet, (g) RoadFormer, (h) TransUNet, (i) TransRoadNet, (j) LightCGS-Net.

edge-scale targets within the background. Our proposed method places greater emphasis on extracting global contextual features, resulting in clearer edge contours compared to existing methods. The integration of high-resolution feature information may contribute to more detailed boundary features. There remains potential for improvement in the segmentation of densely interconnected road structures in urban areas, particularly when compared to ground truth labels. Additionally, we observed that, compared to other transformer-based road networks, the FFM in our design more accurately distinguishes edge and spatial detail features. The fused images, processed through Gaussian filtering, also reduce the interference of redundant information.

Fig. 9 presents a visual comparison chart, consisting of one selected image from each of the cities Las Vegas, Paris, Shanghai, Khartoum, and Moscow in the SpaceNet dataset, showcasing the predicted segmentation performance of fine features in large-scale objects using six different networks. The red and yellow bounding boxes carry the same meaning as in the other two datasets. In comparison with the ground truth images, our network still has potential for improvement in the segmentation of road structural features within densely interconnected urban areas. Compared to other Transformer-based segmentation networks, our network demonstrates superior segmentation performance in capturing surrounding details. When processing complex background information, LightCGS-Net accurately distinguishes small-scale objects from the background, precisely detecting details and edge-scale targets within the background. Additionally, our proposed method effectively balances the extraction of both global contextual features and spatial details, resulting in clearer edge contours than existing road extraction methods. Furthermore, when dealing with occlusions and road blockages, our method avoids segmenting redundant or false roads caused by interference.

5. Conclusion

In this study, we propose LightCGS-Net, a novel dual-branch road feature extraction network for HRSIs. The architecture integrates a lightweight CNN-based spatial detail branch with a Swin Transformer (ST)-based global semantic branch within the decoder, achieving a balance between global semantic understanding and local spatial detail extraction to enhance road segmentation performance. Key innovations include a Multi-Scale Feature Fusion Module (FFM) with dilated convolutions to improve edge detail extraction and a Parallel Channel

Spatial Attention Mechanism (PCSAM) to mitigate road occlusions caused by trees and buildings by refining features in both channel and spatial dimensions. Additionally, the lightweight redesign of residual blocks using depthwise and pointwise convolutions reduces parameters, making the model suitable for edge devices like UAVs. Extensive experiments on benchmark datasets demonstrate the model's superior performance and generalizability, with ablation studies confirming the importance of the ST, FFM, and PCSAM modules. While LightCGS-Net achieves promising results, further optimization is required to improve real-time performance on UAVs, paving the way for advancements in lightweight design without sacrificing segmentation accuracy.

CRedit authorship contribution statement

Yifei Duan: Writing – original draft, Software, Methodology, Conceptualization. **Dan Yang:** Validation, Data curation. **Xiaochen Qu:** Writing – review & editing, Visualization. **Le Zhang:** Investigation. **Junsuo Qu:** Supervision, Project administration, Funding acquisition. **Lu Chao:** Supervision. **Peilu Gan:** Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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